



Public engagement and bioethics as natural allies

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Received: 30 April 2026 / Accepted: 28 May 2026
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Abstract

This article offers an original theoretical basis for the claim that public engagement and bioethics are natural allies. It draws on Walker’s expressive-collaborative model, which conceptualises morality as a social reality in which differently situated individuals continuously negotiate moral understandings and shared responsibilities to sustain social equilibrium. By highlighting the collective, dialogic, and interpretative nature of moral reasoning—and the epistemic partiality of all actors, including experts—it underscores the necessity of intersubjective processes for achieving legitimate and inclusive governance in bioethics. Deliberative forms of engagement materialise this approach by fostering moral knowledge co-production among individuals with unequally distributed authority over moral claims, thereby helping mitigate epistemic power imbalances between laypeople and experts. Realising this democratic ideal, however, requires rethinking engagement practices that are often expert-driven and work in silos (i.e., engaging citizens and experts separately). This limits the potential for mutual learning and the reconfiguration of moral knowledge to reflect the societal values, needs, and concerns. Instead, engagement practices should stress the interdependence and complementarity of laypeople and experts in solving complex ethical issues.

Keywords Bioethics · Public engagement · Genomics · Expressive-collaborative model · Deliberative democracy

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1 Public engagement, a growing trend in bioethics

Despite the absence of consensus on the exact definition of bioethics, it is generally recognised as an interdisciplinary field that examines and provides guidance on ethical issues arising from scientific and technological innovations affecting living beings and their environment (Gordon n.d). In view of the growing impact of science and technologies on society, bioethics has become a “public policy affair” (Schick-tanz, Schweda, and Wynne 2012). In many countries, it is now institutionally organised in ethics committees and expert groups that produce normative mechanisms (e.g., guidelines, recommendations) to influence health policymaking and regulatory frameworks (e.g., laws, professional procedures). Yet, how could bioethics truly be a *public* policy practice without the general public? Already in 1997, Article 28 of the Convention on Human Rights and Biomedicine made explicit the responsibility of Member States to involve citizens in biomedical progress and its applications (Council of Europe 1997). Public engagement aims to raise awareness, cultivate informed discussions, and support policymaking and professional practices by integrating lay perspectives on fundamental moral questions (Ryan and Smith 2014; Saunders 2018). In this context, the societal role of public bioethics bodies gradually emerged, referring to institutions responsible for organising and facilitating public debates and formulating policy recommendations on a democratic basis (Kelly 2003). The trend towards engagement is particularly noticeable in the field of genomics¹, due to its expanding implementation in a wide range of contexts, including medicine, research, public health, and forensics. In the United Kingdom, a national public consultation has advised future newborn screening programs that may be expanded to the entire genome (Hopkins, Kinsella and Evans 2021). In France, citizens have discussed the ethical implications of genetic testing and genomic medicine through an online platform and regional debates, as part of the revision of the bioethical law (Comité Consultatif National d’Éthique 2018). In the Netherlands, public opinion on modifying the DNA of embryos has been collected through 27 offline and online dialogues throughout the country (Gouman, van Baalen and Verhoef 2021). In South Africa, a citizen deliberation has been conducted on genome editing to induce changes to ethical guidelines, legal regulations, and governmental funding for research and clinical care (Thaldar et al. 2022). This engagement ambition can extend beyond national borders to reach a European or international audience, as demonstrated by projects such as SIENNA (Kantar 2019) and ‘Your DNA, Your Say’ (Middleton et al. 2020).

The necessity of addressing ethical matters in genomics through public engagement is now well established, almost unquestionable (Kamenova 2023). Pragmatically, engagement initiatives can improve health policies and healthcare practices by making them more responsive to the actual needs of patients and citizens (Council of Europe 2020; World Health Organization 2024). From a normative angle, engagement ensures the moral accountability of policies and research by incorporating the values, needs, and concerns of affected individuals (Das et al. 2022). Epistemically, laypeople contribute experiential and context-sensitive knowledge that complements

¹ Throughout the paper, we will repeatedly refer to the specific context of genomics to illustrate our arguments, although our analysis equally applies to other bioethical topics.

technical expertise and enhances decision quality (Daudelin et al. 2011). Democratically, engagement strengthens the political system by allowing those with less power to discuss and challenge priorities in agenda setting (Scheinerman 2023).

In response to this excitement, some have questioned the relevance of public engagement in bioethics. Arguing that it rarely leads to consensus and laypeople are ill-equipped to deal with the ethics of scientific topics due to their ignorance and emotional, biased mindset, Rhodes and Ostertag (2023) advocated a return to expertocratic, paternalistic governance in genomics. Likewise, Badano (2012) attempted to demonstrate the weaknesses of most arguments in favour of lay participatory processes, concluding that their only legitimate added value lies in counter-balancing the coercive power of politics. Albeit public engagement has limitations and should not be the sole approach to addressing ethical issues in genomics, this paper seeks to provide more foundational arguments for its meaningful contribution to bioethics. It further develops the statement of Kim et al. (2010) that bioethics and deliberative engagement formats are natural allies. While these authors focused on methodological aspects (comparing the relevance of opinion surveys with deliberative democracy), the present paper offers theoretical grounding—using the expressive-collaborative model of feminist philosopher Walker (2007)—to justify why public engagement and bioethics can be considered natural allies.

2 The expressive-collaborative model of morality

2.1 Morality is interpersonal and empirically constructive

In her book *Moral understandings: A feminist study in ethics*, Margaret Urban Walker (2007) developed a conception of morality as interpersonal and empirically constructive. According to her, morality and ethics are much more than the product of a solitary reflection from a few knowledgeable moral agents, who decide what is right and wrong by weighing abstract moral principles applied to fictional situations. She reminds us of the profoundly human character of all ethical theories and their application to moral life. They are always the product of how particular philosophers, situated in specific social, historical, and cultural contexts, observed particular lived realities, which they interpreted in a specific manner to formulate generic rules to guide humanity. Therefore, as Walker (2007) underlined, canonical ethical theories and their moral rules may not make sense for all moral agents:

Objects of comparison are important because moral philosophers inevitably reflect on morality from within lifeways embodying particular forms of morality, and from a particular position within those lifeways. They may thus find salient only some kinds of understandings and practices and may be completely unaware of others. They may also find interactions intelligible in certain terms from social positions they are familiar with, without knowing that these interactions make different sense to others differently placed, if they make sense to those others at all. (p. 11)

Abstract moral principles acquire their specific meaning and achieve their full potential when used by humans in real situations. Hence, in response to the prevailing tradition of ethics and morality, which Walker depicts as transcendent, intellectualist, and individualist, she proposes an alternative conception grounded in its social and empirical roots. In her view, morality is a complex social reality that moral agents continually construct together through interpersonal interactions. These interactions make people more aware of themselves and others as moral agents, and raise their awareness about their unavoidable interdependence in doing good or bad. This spurs them to negotiate mutual moral responsibilities constantly while also reinforcing reciprocal recognition. To live together harmoniously, humans must create shared intelligibility and agreement regarding what is right and wrong, and who bears responsibilities for what and when. Walker theorised this comprehension of morality through an *expressive-collaborative model*. She argues that morality is something people *express* and build *collaboratively* to make sense of moral understandings for themselves and others. Importantly, the way people develop moral understandings depends on how they perceive and interpret morally relevant elements, such as facts, values, needs, and concerns (Schultz 2009). And the way individuals identify and interpret morally relevant elements depends on their specific position in society, characterised by cultural, socio-economic, and gender norms (Walker 2007). In other words, moral understandings are always situated; they are shaped by people's experience of moral life, and they are negotiated through social interactions, depending on the authority of each protagonist to influence moral norms.

2.2 Different moral understandings

Because people are situated differently in society, it creates different moral understandings that are likely to generate divergent evaluations of bioethical issues. As Walker (2007) emphasised, differently situated people identify different elements as morally relevant, have different interpretations of these elements, and balance those elements differently (e.g., prioritisation of values):

In a society structured in these ways, there are likely to be differences in views about the nature, structure, gravity, and rational resolution of moral problems among people who live different lives. Differently situated people may face different moral problems or experience similar ones differently. They will have reasonably different understandings of costs, risks, and relevance. They will see different responses realistically open to them in responding to these problems, and find different ways of resolving them to be successful or sane (p. 56).

From an epistemic angle, it is questionable whether any single individual—even the most expert and wise one—can grasp the diversity of moral experiences and do so in abstraction from their affinity with some moral understandings. For instance, how could someone healthy, financially secure, and socially protected genuinely imagine how someone living in poverty, who is chronically ill, experiences and prioritises moral values when confronted with an ethical dilemma (Schicktanz et al. 2012)? As Schicktanz et al. (2012) pointed out, bioethical reasoning that fails to engage

with the plurality of perspectives embodied in singular life experiences not only risks overlooking significant insights but also induces justified suspicion of one-sidedness, thereby undermining its legitimacy. Bioethics requires an *intersubjective* process to build moral understandings that are collectively legitimate when dealing with the societal governance of genomic technologies. In this sense, public engagement, and more specifically deliberative formats, can serve as a natural ally to bioethics. Public engagement materialises the expressive-collaborative model of morality, namely the empirically evolving medium by which differently situated people can learn about and negotiate moral understandings, which are sometimes shared and sometimes conflicting, to maintain a social equilibrium.

3 Public engagement embodies the expressive-collaborative model

3.1 A democratic space to co-construct bioethics

Deliberative methods convene various individuals or stakeholders to discuss issues in depth, consider multiple viewpoints, and generate collective input. Well-known examples include citizen juries, assemblies, and forums (Marien and Felicetti, 2019; Smith and Wales 2000). The engagement format is commonly structured around three steps. The first step involves providing equal access to high-quality information for all participants, allowing them to start the conversation on a shared contextualisation of the issues at stake. The second step is to create a discussion climate that stimulates critical thinking and mutual learning. In the final step, participants may collectively produce recommendations on what matters ethically and translate these into actions that sustain common interests (Abelson et al. 2003). When following these standards, deliberative formats provide a safe and democratically structured environment for differently situated individuals, with unequally distributed authority in moral claims, to co-construct moral issues. Such an egalitarian space encourages participants to listen to each other and express what they value. Sharing experiences in profoundly human and relational matters creates the right conditions to become more open towards the moral comprehension of others living in distinct realities. This democratic configuration creates fertile ground for reinterpreting the present of moral issues and collaboratively building a future based on a more representative view of the moral norms at play in society.

In other words, public engagement contributes to better bioethics by fulfilling a reflective and a normative role simultaneously. It invites participants to critically reflect together on the moral norms at stake and their various interpretations from different standpoints. The value of critical reflection, in Walker's (2007) eyes, lies in verifying whether moral understandings are intelligible and coherent for those performing them in real-life situations, but from different positions. The lack of intelligibility or coherence manifests the urge to revise moral norms (e.g., their meaning or prioritisation) by seeking shared agreement where possible and, otherwise, at least clarifying irreconcilable disagreements (Walker 2007). Creating a common space for deliberation that challenges participants' viewpoints and preconceptions is all the more important considering the natural propensity of human minds to seek informa-

tion that confirms their own beliefs (Kim et al. 2010). Even though interdisciplinary expert bodies are already better at solving issues than solitary reflection, they still represent a prevailing group, sharing some moral intuitions from the outset, and hence, they will miss the social reality of most patients and citizens. The more heterogeneous a group is, the greater its diversity bonus increases, namely its capacity to think outside the box (Page 2017). Engaging with patients can change clinicians' and health policymakers' understanding of a disease, shifting their knowledge and attitudes about what kinds of policies and healthcare practices are judged appropriate (World Health Organization 2024). For example, involving citizens in a DNA Debate has provided new interpretations and types of vulnerability in genomics that had been omitted in the scientific and ethical literature so far (Mayeur et al. 2023). These vulnerabilities, as expressed by laypeople, are critical to consider in the ethical governance of technologies because they apply to everyone confronted with genomics, not solely to particularly vulnerable groups. Especially when the issue under discussion involves a risk assessment, which is heavily contingent on stakeholders' social conditions, the moral understandings of diversely situated patients and citizens should not be overlooked (Scheinerman 2023).

Given the rapid pace at which genomic technologies are advancing, and the new ethical dilemmas they raise, it is increasingly significant to recognise the empirical, evolving, and pluralistic essence of moral reasoning. Morally relevant elements must be continually reassessed in light of unprecedented, concrete situations and through dialogue with those who experience them in diverse ways. Indeed, it is precisely through deliberation on new and complex cases that we gain the deepest insight into both the present state and the desirable future of morality (Richardson 2018). Establishing ethical standards based on the lived realities of affected individuals does not mean that bioethics relies on subjective opinions, but that it considers what differently situated stakeholders truly value and have legitimate reasons to defend (Schneider et al. 2023).

3.2 Managing conflicts caused by different moral understandings

Another valuable function of public engagement as the natural ally of bioethics lies in its potential to manage conflicts caused by inevitable differences in moral understandings. As previously highlighted, the diversity of moral agents' experiences in ethically sensitive situations, partially determined by their specific positions in society, generates divergent moral evaluations of the issues at stake and their appropriate resolution. Such a plurality frequently leads to controversies, disagreements, and misunderstandings. Yet, addressing moral conflicts should not necessarily be equated with pursuing a consensus, a narrow conception of public engagement that has faced increasing criticism (Kamenova 2023; Middleton et al. 2023). Achieving consensus on bioethical topics is arduous and often unattainable, given their contentious nature and the range of underlying values of differently situated people. This limitation, nonetheless, does not render public engagement ineffective for bioethics, contrary to what Rhodes and Ostertag (2023) asserted. Rather, its value lies in a more elaborate repertoire of conflict-management functions.

Constructive dialogues help pinpoint and mitigate misunderstandings among people who inhabit different social realities to counter the polarisation of opinions. Evidence from deliberative practices shows that participants are generally willing to learn from one another and remain open to changing their views, as well as that the reciprocal exchange of intimate experiences enables them to grow together by rethinking their prejudices, cultivating tolerance, and creating new types of knowledge that did not exist before the engagement (Abma et al. 2010; Grönlund et al. 2015; Houtman et al. 2023). By moving beyond an adversarial mindset of persuading others of their own correctness, participants tend to develop a shared vision that serves the common interest (Meagher and Lee 2016). Such collective effort prompts participants to step back from first-order contentious discourses to examine the underlying values, needs, and concerns that, ultimately, are the fundamentals to be discussed to preserve social equilibrium. This is how deliberative engagement facilitates the identification of shared values that transcend stakeholders' particular interests and agendas, while simultaneously providing a space for negotiating moral tensions. On this point, it is worth noting a social phenomenon reported by Felt et al. (2009) in their engagement exercise, convening citizens and genomics researchers. They observed that participants from both groups may strategically withhold controversial issues when they perceive them as potentially too threatening to the social setting. Facilitators are crucial in this context, as they encourage participants to openly discuss conflicting viewpoints. As Walker (2007) pointed out, "Imperfect understandings, conflicting judgements, or incomprehension are [...] opportunities to rethink understandings or search for mediating ideas or reconciling procedures within or between communities" (p. 78). In cases of inextricable moral disagreements, List and Pettit (2011) demonstrated that participants will inevitably have incentives to alter their individual preferences to attain an outcome optimal for the collectivity. It is only here that the consensus-building function of public engagement comes into play, and not as a primary goal. Citizen assemblies have proven to be a notably effective method in this regard, as demonstrated by the Irish initiative that led to a change in abortion law despite a political stalemate of several years (Council of Europe 2020).

The ultimate aim of an expressive-collaborative model is to preserve the fragile social equilibrium among interdependent stakeholders, including patients, healthcare professionals, researchers, and health policymakers, who must collaborate to guarantee the effective functioning of society. To some extent, they all share a common ambition, such as improving medicine through genomics, even though their needs and priorities differ regarding how to achieve this. Since conflicting perspectives are ineluctable, they should be managed within a safe, democratic, and respectful environment to foster fruitful discussions. In a nutshell, public engagement should be conceptualised as a mutually beneficial process in which participants gain self-learning, collective reflection, and community support. This process nourishes the interpersonal dimension of morality at both individual and collective levels. On the one hand, it nurtures participants' sense of autonomy by allowing them to contribute to decisions affecting them. On the other hand, it counteracts social fragmentation by reconnecting individuals and communities, such as patients and policymakers, who would otherwise remain distant. In its expressive-collaborative form, public engagement humanises bioethics by reminding us of our interdependence in solving com-

plex ethical dilemmas and the consequential need for mutual recognition, respect, and care.

4 Deconstructing epistemic power imbalances

By explaining how morality operates within a social environment structured around power dynamics, the expressive-collaborative model of Walker (2007) enables us to grasp the societal implications of how bioethics is performed and by whom. Those who practice ethics or have the authority to shape ethics actually determine which moral norms are dominant and which should be ignored. This is problematic since morality is not the privilege of a few specialists but belongs to and impacts everyone. No one can escape morality, for it is an integral part of life and the human condition, as Vaughn (2010) summarised it well:

Morality is part of the unavoidable, bittersweet drama of being persons who think and feel and choose. [...] Whether we like it or not, we seem confronted continually with the necessity to deliberate about right and wrong, to judge someone morally good or bad, to agree or disagree with the moral pronouncements of others, to accept or reject the moral outlook of our culture or community, and even to doubt or affirm the existence or nature of moral concepts themselves. Moral issues are thus inescapable—including (or especially) those that are the focus of bioethics. (p. 3)

4.1 Bioethics is not the prerogative of people with expertise

Scientific expertise enjoys a high status as factual knowledge serves to anticipate and control uncertainties, which are seen as impediments to the efficient functioning of our society. However, the epistemic authority of scientists to describe how the world functions does not mean they can decide alone how the world *should* function (Schicktanz et al. 2012). Since its origins, bioethics has adopted a critical perspective on the relationship between medical expertise and the legitimacy to determine what is right and wrong for all (Abma et al. 2010). Medical paternalism has lost ground as patients' rights have gained some, and patients have been increasingly recognised as autonomous and responsible agents in their health. Yet, to remain honest and self-reflective, bioethics should apply the same sceptical approach to ethicists. While acknowledging the ongoing debate on whether or not moral experts exist and whether ethicists are moral experts or not, and without committing to any bold claims on this matter, we rely once more on Walker's expressive-collaborative model to argue that in any case, ethicists should not be allowed to dictate moral behaviours based solely on abstract principles and normative theories, disconnected from the people to whom the moral rules apply (Archard 2011; Gordon 2023; Levitt 2003; Musschenga 2009). Abma et al. (2010) outlined a historical transition in bioethics, moving from a dominance of theoretical rationality, in which ethical expertise was decontextualised, to a recognition of practical rationality, in which knowledge is context-specific and, con-

sequently, patients and citizens acquire epistemic legitimacy when expressing and making sense of their moral experiences and underlying values, needs, and concerns.

By integrating citizens' perspectives alongside those of experts in ethics, science, and policymaking, who traditionally govern genomic applications and their ethical dilemmas, public engagement contributes to a better balance in who represents bioethics. This way, it also helps reduce epistemic injustice. Epistemic injustice denotes the systematic underestimation, misrepresentation, or exclusion of some individuals' or groups' knowledge or narratives, in contrast to the excessive credit given to others, typically the socially most privileged ones (Fricker 2007). Such injustice can have far-reaching repercussions, most notably the strengthening of systemic inequalities and biases (Dotson 2011; Medina 2013) and the erosion of public trust (Parkhurst 2017). The Council of Europe (2020) recalled that public engagement must challenge experts' assumptions by opening up debate since the way issues are framed underpins certain values and, therefore, renders others invisible. It would be unfair to allow the values of a minority of experts to dominate the governance of genomics, while disregarding the majority of its technological end-users and data providers. In particular, engaging vulnerable groups and minorities must be a priority to reduce epistemic inequalities. Reaching these populations is one of the main weaknesses of engagement practices and, therefore, one of the greatest challenges to overcome.

Nonetheless, the intention is not to move from an expertocratic approach to a lay dictatorship in bioethics, where everyone would have the same epistemic authority. What matters is recognising the interdependence and complementarity of scientists, ethicists, and laypeople to make bioethics more fair, sustainable, and trustworthy, and the necessity of rethinking engagement practices accordingly.

4.2 Rethinking engagement practices to reduce epistemic power imbalances

Historically, public engagement first emerged in the form of what has been called the deficit model, which consisted of an unilateral transfer of knowledge from scientists to the public with the intention of restoring trust in and enthusiasm for science (Wynne 1991). Within this model, only experts were considered rational and knowledgeable agents, whereas laypeople were seen as rather ignorant and emotion-driven (Felt et al. 2009; Wynne 2006). Despite undeniable progress in mentalities, including the rising interest of the scientific community for mutual learning activities (Boon et al. 2022), many initiatives tend to perpetuate the dichotomy between experts and laypeople, instead of focusing on their complementarity in moral knowledge co-production.

Public engagement frequently operates in silos and has little impact on those who shape moral issues, whether they are scientists, ethicists, or health policymakers (Wirz et al. 2020). Constructive exchanges traditionally occur only among citizens, restricting interactions with experts to educational sessions and with policymakers to the delivery of final recommendations, without subsequent follow-up on their implementation. As a consequence, the role of laypeople is often confined to sharing their personal experiences, whilst scientists are expected to provide technical and 'objective' knowledge, and ethicists to offer external, theoretical normative frameworks. Nevertheless, these traditional roles in participatory processes should not be

considered absolute: patients may possess expertise in their own diseases, and scientists and ethicists remain subjective and emotional beings behind their accumulated knowledge and skills. In a case study illustrating the concept of “inter-ethics”, where patients, healthcare professionals, researchers, and policymakers engaged in a genuine, non-hierarchical dialogue, Abma et al. (2010) observed that the predefined roles and identities of each stakeholder were deconstructed to leave more room for increased awareness and appreciation of their interdependence to improve medical practices. If the ambition is to make the expressive-collaborative model a reality, laypeople should be involved on an equal footing with scientists and policymakers in a common dialogical activity. Ideally, those who lack moral authority should be given the floor first, allowing them to influence the issues from the outset (Abma et al. 2010). Indeed, the points of view of influential actors already dominate outside the engagement setting, while the latter may be the only opportunity for patients and citizens to contribute to issues impacting them (Mayeur et al. 2025). Furthermore, one should take into account some lessons learned from previous observational studies in engagement settings (Daudelin et al. 2011; Felt et al. 2009): participants, whether they are laypeople or experts, need moments of discussion amongst peers to feel comfortable expressing what they truly value and think, without fearing judgment from people with different backgrounds and interests. Abma et al. (2010) adopted this approach in their inter-ethics case study, where the issues were first discussed in peer subgroups to allow each participant to develop their own moral understanding before negotiating them in heterogeneous exchanges.

Finally, an expressive-collaborative model is not simply the addition of varied knowledge and insights, but their fusion, which reconfigures what was known and considered morally right so far. A first condition for new types of knowledge to emerge from a collective process is that each party recognises and trusts the added value and legitimacy of the others. This requires, in particular, that influential actors make additional efforts to question themselves and acknowledge that their expertise does not provide them with omniscient knowledge. A second condition is flexibility in the engagement design to give participants the opportunity to articulate perspectives and issues that are often overlooked in more directive, agenda-driven approaches (Mayeur et al. 2025).

It is important to note that the present paper does not advocate for public engagement as the only legitimate approach in bioethics. Rather, it attempts to demonstrate why it is an essential approach, which complements ethics committees and expert groups.

5 Conclusion

As science and technology increasingly shape our lives, bioethics has become a societal practice. Public engagement now plays a growing role in informing health policies, research, and clinical care. Yet, the faith in lay involvement with complex bioethical issues raises questions about the legitimacy of their contributions. Drawing on Walker’s expressive-collaborative model (2007), this article proposes an original theoretical framework to demonstrate why and how public engagement and

bioethics can be considered natural allies. This model views morality as a social reality that individuals continuously express and build together. Through their interactions, moral agents continually negotiate moral understandings and responsibilities to maintain social equilibrium. However, individuals with diverse social positions and epistemic authority tend to develop distinct moral understandings. They often identify diverse elements as morally relevant and interpret or prioritise these elements differently. By highlighting the collective, dialogic, and interpretative nature of moral reasoning—and the epistemic partiality of all actors, including experts—it underscores that intersubjective processes are necessary to achieve legitimate and inclusive governance in bioethics.

In particular, deliberative engagement formats bring together diverse individuals to foster mutual learning, reciprocal moral recognition, and the negotiation of moral understandings in a safe and democratic space. This enables the co-production of moral knowledge based on a greater diversity of experiences, redefining what was previously understood and admitted as morally right. By enabling more egalitarian dialogues between stakeholders who would otherwise never meet, such as patients and health policymakers, those participatory processes mitigate social conflicts and polarisation. Additionally, such an environment enhances participants' moral reasoning by stimulating self-reflectivity, open-mindedness, empathy, and collective visions. Finally, public engagement helps reduce epistemic power imbalances by integrating lay perspectives alongside those of experts in science, ethics, and policymaking.

Author contributions C.M. wrote the drafts and final version of the manuscript, while W.VH., M.S., and H.M. provided fundamental reviews and contributed to its final structure and content. All authors approved the final manuscript.

Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests All authors are regularly involved in public engagement projects and receive funding from various sources, including European institutions and national agencies.

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References

- Abelson, Julia et al. 2003. Deliberations about deliberative methods: Issues in the design and evaluation of public participation processes. *Social Science & Medicine* 57(2):239–251. [https://doi.org/10.1016/S0277-9536\(02\)00343-X](https://doi.org/10.1016/S0277-9536(02)00343-X)

- Abma, Tineke, Baur, Vivianne, Molewijk, Bert, and Widdershoven, Guy. 2010. Inter-ethics: Towards an interactive and interdependent bioethics. *Bioethics* 24:242–255. <https://doi.org/10.1111/j.1467-8519.2010.01810.x>
- Archard, David. 2011. Why moral philosophers are not and should not be moral experts. *Bioethics* 25 (3): 119–127. <https://doi.org/10.1111/j.1467-8519.2009.01748>
- Badano, Gabriele. 2012. Genomics and public involvement: Giving justifications their due. *Studies in Ethics Law and Technology* 6(1). <https://doi.org/10.1515/1941-6008.1176>
- Boon, Wouter et al. 2022. Meaningful public engagement in the context of open science: Reflections from early and mid-career academics. *Research for All* 6(1):23. <https://doi.org/10.5281/zenodo.7304239>
- Comité Consultatif National d'Éthique. 2018. Rapport de synthèse du Comité Consultatif National d'Éthique: Opinions du comité citoyen. EDP Sciences. <https://www.vie-publique.fr/files/rapport/pdf/184000352.pdf>. Accessed 25 March 2026.
- Council of Europe. 1997. Convention for the protection of human rights and dignity of the human being with regard to the application of biology and medicine: Convention on human rights and biomedicine. <https://rm.coe.int/168007cf98>. Accessed 25 March 2026.
- Council of Europe. 2020. Guide to public debate on human rights and biomedicine. <https://rm.coe.int/prems-009521-ex-061320-gbr-2007-guide-on-public-debate-16x24-web/1680a12679>. Accessed 25 March 2026.
- Das, Jayatri et al. 2022. Neuroscience is ready for neuroethics engagement. *Frontiers in Communication* 7(909964). <https://doi.org/10.3389/fcomm.2022.909964>
- Daudelin, Geneviève, Lehoux, Pascale, Abelson, Julia, and Denis, Jean L. 2011. The integration of citizens into a science/policy network in genetics: Governance arrangements and asymmetry in expertise. *Health Expectations* 14(3):261–271. <https://doi.org/10.1111/j.1369-7625.2010.00636.x>
- Dotson, Kristie. 2011. Tracking epistemic violence, tracking practices of silencing. *Hypatia* 26(2):236–257. <https://doi.org/10.1111/j.1527-2001.2011.01177.x>
- Felt, Ulrike, Fochler, Maximilian, Müller, Annina, and Strassnig, Michael. 2009. Unruly ethics: On the difficulties of a bottom-up approach to ethics in the field of genomics. *Public Understanding of Science* 18(3):354–371. <https://doi.org/10.1177/0963662507079902>
- Fricker, Miranda. 2007. *Epistemic injustice: Power and the ethics of knowing*. Oxford: Oxford University Press.
- Gordon, John-Stewart . No date. Bioethics. Internet Encyclopedia of Philosophy. <https://iep.utm.edu/bioethics/#H1>
- Gordon, John-Stewart. 2023. Moral expertise revisited. *Bioethics* 37:533–542. <https://doi.org/10.1111/bioe.13172>
- Grönlund, Kimmo, Herne, Kaisa, and Setälä, Maija . 2015. Does enclave deliberation polarize opinions? *Political Behavior* 37(4):995–1020. <https://doi.org/10.1007/s11109-015-9304-x>
- Hopkins, Henrietta, Kinsella, Suzannah, and Evans, Grace . 2021. Implications of whole genome sequencing for newborn screening: A public dialogue. Hopkins Van Mil. Accessed 25 March 2026. <https://www.genomicsengland.co.uk/assets/documents/public-dialogue-wgs-for-nbs-final-report.pdf>
- Houtman, Diewertje et al. 2023. Changes in opinions about human germline gene editing as a result of the Dutch DNA-dialogue project. *European Journal of Human Genetics* 31:409–416. <https://doi.org/10.1038/s41431-022-01114-w>
- Kamenova, Kalina. 2023. The intrinsic value of public deliberation in the governance of human genome editing. *The American Journal of Bioethics* 23(7):63–65. <https://doi.org/10.1080/15265161.2023.2215608>
- Kantar. 2019. SIENNA D2.6: Qualitative research exploring public attitudes to human genomics. Zenodo. <https://zenodo.org/records/4081193>. Accessed 26 March 2026.
- Kelly, Susan E. 2003. Public bioethics and publics: Consensus, boundaries, and participation in biomedical science policy. *Science Technology and Human Values* 28(3):339–364. <https://doi.org/10.1177/0162243903028003001>
- Kim, Scott Y.H., Wall, Ian F., Stanczyk, Aimee, and De Vries, Raymond. 2010. Assessing the public's views in research ethics controversies: Deliberative democracy and bioethics as natural allies. *Journal of Empirical Research on Human Research Ethics* 4(4):3–16. <https://doi.org/10.1525/jer.2009.4.4.3>
- Levitt, Mairi. 2003. Public consultation in bioethics: What's the point of asking the public when they have neither scientific nor ethical expertise? *Health Care Analysis* 11(1):15–25. <https://doi.org/10.1023/A:1025381828650>
- List, Christian and Philip Pettit. 2011. *Group agency: The possibility, design, and status of corporate agents*. Oxford: Oxford University Press.

- Marien, Sofie and Felicetti, Andrea. 2019. Citizen forum on the use of genome information in health care: An assessment of the quality of the process. KU Leuven. Accessed 30 March 2026. <https://soc.kuleuven.be/afdeling-politieke-wetenschappen/centre-for-political-research/legit/files/citizens-reporten-executive-reportnl-fr.pdf>
- Mayeur, Chloé, Howard, Heidi C., and Van Hoof, Wannes. 2025. A citizen-centred approach to public engagement on the ethical, legal and societal issues of health technologies. *Public Understanding of Science* 34(7):901–915. <https://doi.org/10.1177/09636625251326490>
- Mayeur, Chloé, Mertes, Heidi, and Van Hoof, Wannes. 2023. Do genomic passports leave us more vulnerable or less vulnerable? Perspectives from an online citizen engagement. *Humanities and Social Sciences Communications* 10:1467. <https://doi.org/10.1057/s41599-023-01580-7>
- Meagher, Karen M., and Lee, Lisa M. 2016. Integrating public health and deliberative public bioethics: Lessons from the human genome project ethical, legal, and social implications program. *Public Health Reports* 131(1):44–51. <https://doi.org/10.1177/003335491613100110>
- Medina, José. 2013. *The epistemology of resistance: Gender and racial oppression, epistemic injustice, and the social imagination*. Oxford University Press. <https://doi.org/10.1093/acprof:oso/9780199929023.001.0001>
- Middleton, Anna et al. 2020. Members of the public in the USA, UK, Canada and Australia expressing genetic exceptionalism say they are more willing to donate genomic data. *European Journal of Human Genetics* 28:424–434. <https://doi.org/10.1038/s41431-019-0550-y>
- Middleton, Anna et al. 2023. Public engagement with genomics. *Wellcome Open Research* 8:310. <https://doi.org/10.12688/wellcomeopenres.19473.2>
- Musschenga, Albert. 2009. Moral intuitions, moral expertise and moral reasoning. *Journal of philosophy of education* 43(4):597–613.
- Page, Scott E. 2017. *The diversity bonus: How great teams pay off in the knowledge economy*. Princeton: Princeton University Press.
- Parkhurst, Justin. 2017. *The politics of evidence: From evidence-based policy to the good governance of evidence*. Abingdon: Routledge.
- Rhodes, Rosamond, and Ostertag, Gary . 2023. Public engagement in shaping bioethics policy: Reasons for skepticism. *The American Journal of Bioethics* 23(7):72–74. <https://doi.org/10.1080/15265161.2023.2207505>
- Richardson, Henry S. 2018. Moral reasoning. The Stanford Encyclopedia of Philosophy. <https://plato.stanford.edu/archives/fall2018/entries/reasoning-moral/>. Accessed 25 March 2026.
- Ryan, Matthew, and Smith, Graham . 2014. Defining mini-publics. In *Deliberative mini-publics: Involving citizens in the democratic process*, eds. Grönlund Kimmo, Bächtiger André, and Setälä Maija, 9–26. Colchester: ECPR Press.
- Saunders, Tom. 2018. Seven principles for public engagement in science and innovation policymaking: A guide from Nesta's inclusive innovation team. Nesta. https://media.nesta.org.uk/documents/Seven_principles_HILwdow.pdf. Accessed 25 March 2026.
- Scheinerman, Naomi. 2023. Public engagement through inclusive deliberation: The human genome international commission and citizens' juries. *American Journal of Bioethics* 23(12):66–76. <https://doi.org/10.1080/15265161.2022.2146786>
- Schicktan, Silke, Schweda, Mark, and Wynne? Brian . 2012. The ethics of 'public understanding of ethics'—Why and how bioethics expertise should include public and patients' voices. *Medicine, Health Care and Philosophy* 15:129–139. <https://doi.org/10.1007/s11019-011-9321-4>
- Schneider, Manuel, Vayena, Effy, and Blasimme, Alessandro . 2023. Digital bioethics: Introducing new methods for the study of bioethical issues. *Journal of Medical Ethics* 49(11):783–790. <https://doi.org/10.1136/medethics-2021-107387>
- Schultz, Dawson S. 2009. Hurrah for empirical bioethics (where hermeneutically clarified) or how perception of facts 'depends' on values. *The American Journal of Bioethics* 9(6–7):95–99. <https://doi.org/10.1080/15265160902918655>
- Smith, Graham, and Wales, Corinne. 2000. Citizens' juries and deliberative democracy. *Political studies* 48:51–65. <https://doi.org/10.1111/1467-9248.00250>
- Thaldar, Donrich et al. 2022. A deliberative public engagement study on heritable human genome editing among South Africans: Study results. *PLOS One* 17(11):e0275372. <https://doi.org/10.1371/journal.pone.0275372>
- Gouman, Jeroen, van Baalen, Sophie, and Verhoef, Petra. 2021. Resultaten van de DNA-dialoog : Zo denken Nederlanders over het aanpassen van embryo-DNA. DNA-dialoog <https://dnadialoog.nl/wp-content/uploads/2021/01/Eindrapport-DNA-dialoog-jan-2021.pdf>

- Vaughn, Lewis. 2010. Moral reasoning in bioethics. In *Bioethics principles, issues, and cases*, New-York: Oxford University Press.
- Walker, Margaret U. 2007. *Moral understandings: A feminist study in ethics* (second edition). New-York: Oxford University Press.
- Wirz, Christopher D., Scheufele, Dietram A., and Brossard, Dominique . 2020. Societal debates about emerging genetic technologies: Toward a science of public engagement. *Environmental Communication* 14(7):859–864. <https://doi.org/10.1080/17524032.2020.1811478>
- World Health Organization. 2024. *Citizen engagement in evidence-informed policy-making: A guide to mini-publics*. Geneva: World Health Organization.
- Wynne, Brian. 1991. Knowledges in context. *Science Technology and Human Values* 16(1):111–121. <https://doi.org/10.1177/016224399101600108>
- Wynne, Brian. 2006. Public engagement as a means of restoring public trust in science: Hitting the notes, but missing the music? *Community Genetics* 9(3):211–220. <https://doi.org/10.1159/000092659>

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